

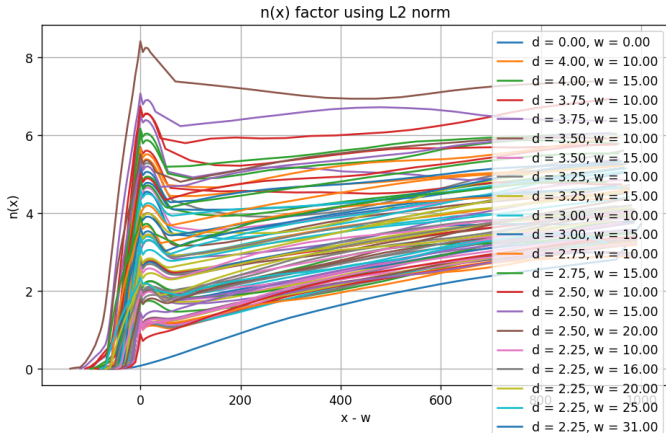
IMPERIAL

Finishing incNS and starting comNS

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September 18, 2025

n-factors

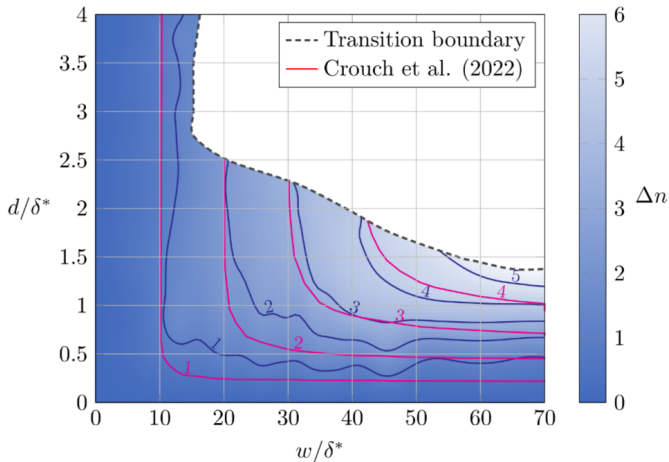
- Different behaviour in the evolution of the n-factors after the gap. Why? Maybe the unstable frequencies are damp out on the growth in the gap?



Δn -factors

Steps to compute Δn (after introducing some noise as body force in the upstream part of the domain):

- Δn averaged in the window (gap lies in $x \in (0, w)$):
 $x \in (w + 50, w + 250)$.
- Cubic interpolation inside the convex hull.
- Gaussian filter with $\sigma = 2$. Maybe too much? I was getting some weird stripes without filter.



Compressible Simulations

So far they ran, but I had a problem with the blasius that was triggering noise.

- Polynomial order?
- CFL? with the implicit solver. 2 is ok? This translates to $dt \sim 0.06$